Automated Rice Leaf Disease Diagnosis Using CNNs

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***Abstract*—Rice is a staple food in Bharat (India) and many other parts of the world. However, the increasing demand for rice due to population growth forces various challenges, including degraded crop quality and quantity due to rice plant diseases. Diseases such as brown spots, bacterial blight, and hispa can significantly reduce farming output, thereby impacting the productivity of the agriculture sector. To address this challenge, various solutions such as Agricultural cyber-physical systems (ACPS) and precision agriculture have been proposed, along with the application of deep learning techniques. This paper presents a rice leaf disease detection method using deep transfer learning. The proposed approach explores well-known pre-trained deep Convolutional Neural Network (CNN) models - VGG19, DenseNet201, InceptionV3, ResNet50, EfficientNetB3, EfficientNetB7, and XceptionNet, for image-based rice disease classification. Experimental results show that the DenseNet model by the proposed method achieved the highest classification accuracy of 98.75% when fine-tuned properly. The proposed scheme outperforms many existing approaches, delivering a superior disease control solution for rice leaf diseases.**

***Index Terms*—Rice Cultivation; Rice Diseases Detection; Agriculture Cyeber-Physical System; Deep Learning Techniques; Performance Metrics.**

I. Introduction

Agriculture is a crucial contributor to the economic growth of any country. However, with the increase in population, meeting food demands has become challenging. To tackle this challenge, innovative farming techniques based on cyber-physical systems (CPS) have emerged as a cutting-edge solution for sustainable agriculture, commonly known as precision agriculture [1]. Precision agriculture involves the integration of digital technologies and data-driven farm management practices, leveraging the capabilities of the Internet of Things (IoT) in agriculture. This approach optimizes farming processes such as planting, fertilizing, watering, and harvesting using sensors, data analytics, and automation.

Rice is one of the most important cereal crops in the world, providing food for more than half of the global population [2]. Today, rice is a staple food for the majority of the population in Bharat, with millions of people depending on it as their primary source of nutrition and sustenance. Rice consumption in Bharat is also deeply connected to religious and cultural practices. For example, rice is a standard offering in Hindu temples, and rice-based dishes such as biryani, pulao, and idli are popular in different parts of the country. Overall, rice is an indispensable crop in Bharat, vital to food security, cultural practices, and the economy. In Bharat, rice is cultivated in almost every state, and the country is the second largest rice producer after China. However, rice crops are vulnerable to diseases caused by bacteria, fungi, viruses, and other pathogens. For example, some of the most common rice diseases are bacterial leaf blight, bacterial leaf streak, blast, panicle blight, brown spot, dead heart, downy mildew, hispa, and tungro. If these diseases are not treated on time, they can significantly reduce crop yield, quality, and profitability and threaten food security.

Developing deep learning methods, especially convolutional neural networks (CNN), has made significant progress over the past few decades [3], transforming every sector of the global economy, including agriculture. CNN has been created as a potent tool to help farmers monitor and detect plant diseases in response to demands for more significant crop output. Many agricultural researchers [4], [5], [6] are studying methods for detecting rice diseases that can help farmers make more informed decisions. Although the current study in the state-of-the-art including [2], [7] uses different deep learning models, a robust methodology is not guaranteed due to the small dataset. One approach to preventing crop loss due to disease infection is to capture images of diseased leaves and analyze them to gain insight into the ailment. The proposed device would detect the disease that has developed automatically.

This paper develops an early diagnosing method for rice plant diseases, specifically leaf diseases. The proposed method employs a few well-performed pre-trained CNN models such as VGG19, DenseNet201, etc. The selected models are trained on a larger dataset containing 14416 images to classify rice leaf disease correctly. Experimental results achieve adequate results in these disease detection. The highest training and test accuracy reach 99.99% and 98.75%, respectively, which are pretty satisfactory to many existing methods. Thus, the model's outcome can be an advisory and an early detection tool in agriculture, helping farmers identify and manage rice plant diseases effectively.

The rest of this paper is organized as follows. Section II presents related works. Section III describes the proposed approach. Section IV provides experimental results. Section V concludes the paper.

II. Related Works

Cyber-physical systems in agriculture have been revolutionized by introducing deep learning techniques, frequently

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deployed to address diverse challenges with greater efficacy than conventional approaches. A notable application of these techniques has been in detecting rice plant diseases, where researchers have made substantial progress, surpassing previous limitations. For example, Chen *et al.* [2] used different deep learning models to identify rice diseases. Their experimental results show the highest validation accuracy of 99.25% with Mobilenet architecture. Yang *et al.* [8] provided a deep convolutional neural network to classify the ten common rice diseases. Their experiment includes the strategy of 10-fold cross-validation. The process achieved the highest validation accuracy of 95%. Jayareka *et al.* [9] presented a deep learning model to classify rice diseases into four categories, i.e., rice blast, bacterial leaf blight, sheath blight, and healthy leaves. The procedure is based on converting the original RGB images into HSV images to remove the background noise and use the modified InceptionV3 deep learning model later. Their study shows an accuracy of 91.9% for the blast affected, 93.2% for the bacterial blight, 89% for the sheath rot, and 90% for the brown spot. Vydeki *et al.* [10] optimized a deep neural network for classifying rice diseases. The procedure is based on image segmentation, in which the k-means clustering method is applied to segment the normal and diseased portions. Their model evaluation shows the highest accuracy of 98.9% for the blast affected, 95.78% for the bacterial blight, 92% for the sheath rot, 94% for the brown spot, and 90.57% for the typical leaf image. Lin *et al.* [11] developed a RiceTalk application that uses AI mechanisms and IoT devices to recognize rice blast diseases. Their method achieved an accuracy of 89.4%. Mahesh *et al.* [12] offered a rice disease detection mechanism using the transfer learning technique. The process involved rigorous experiments on pre-trained models. The VGG16 model produced the best classification accuracy of 93.11%. Arnab *et al.* [13] used a CNN architecture using inception and residual connection. The architecture achieved an accuracy of 99.66% on evaluation. The rice disease recognition methods study described above has proven unsatisfactory, making processes less robust and incomplete. Consequently, there is a need for an alternative approach to improve several existing procedures.

III. PROPOSED WORK

This section introduces a customized deep learning method for early rice leaf disease identification. The proposed method can identify nine types of rice leaf disease and ensure the rice plant's health. A deep learning model is a mathematical model with several parameters that must be learned from the data. These parameters represent critical characteristics of the model, such as its complexity or learning rate. We first provide the hyperparameter tuning and then describe the disease detection mechanism.

A. Hyperparameter Tuning

Hyperparameter tuning selects a deep learning model's optimal combination of hyperparameters to maximize its performance on a specific task. Hyperparameters are adjustable parameters not learned by the model during training but are set before the training process. They are often used to control the complexity of the model, such as the number of hidden layers in a neural network, learning rate, regularization strength, or batch size. Hyperparameter tuning is crucial to achieving the best possible performance of a deep learning model. The following hyperparameters are fine-tuned in the proposed methodology.

1) Learning Rate: The learning rate is one of the most critical hyperparameters for optimizing Neural Network performance. It controls how much the model parameters are adjusted concerning the gradient of the loss function. A high learning rate may cause the optimization algorithm to overshoot the minimum of the loss function. In contrast, a low learning rate may result in a very slow convergence to the minimum. Therefore, a good choice of learning rate is crucial for achieving optimal model performance. We propose a scheduler described in Algorithm 1 to achieve an optimal learning rate based on training accuracy and validation loss. The algorithm updates the learning rate after every epoch to get optimal results.

Algorithm 1 Learning Rate Scheduler

Input: Learning rate (lr), Threshold value(th) and epoch number

Output: Optimal Learning Rate (lr)

$lr \leftarrow 0.001$
$th \leftarrow 0.9$
if training accuracy $(tr_acc) < th$ **then**
  if $tr_acc > prev_tr_acc$ **then**
    adjust lr based on tr_acc
  else
    $lr \leftarrow lr * 0.5$
  end if
else
  if validation loss $(val_loss) < th$ **then**
    adjust lr based on val_loss
  else
    $lr \leftarrow lr * 0.5$
  end if
end if

2) Dropout: Dropout is a regularization technique used in deep learning models to prevent overfitting. During training, it randomly drops some neurons out (i.e., setting them to zero). This forces the remaining neurons to learn more robust and independent representations, making the model less sensitive to noise and reducing overfitting. Dropout is controlled by a hyperparameter called the "dropout rate", which specifies the fraction of neurons to be dropped out. The proposed approach shows adequate performance at the optimal dropout rate of 0.45.

3) Regularization: Regularization involves adding a penalty term to the loss function during training. There are two common forms of regularization, namely L1 and L2 regularization, which penalize large weights and encourage sparsity. L1 regularization is achieved by adding the sum of the absolute values of the weights to the loss function. On the other hand, overfitting is prevented by L2 regularization by penalizing the squared magnitude of the weight parameters to the loss

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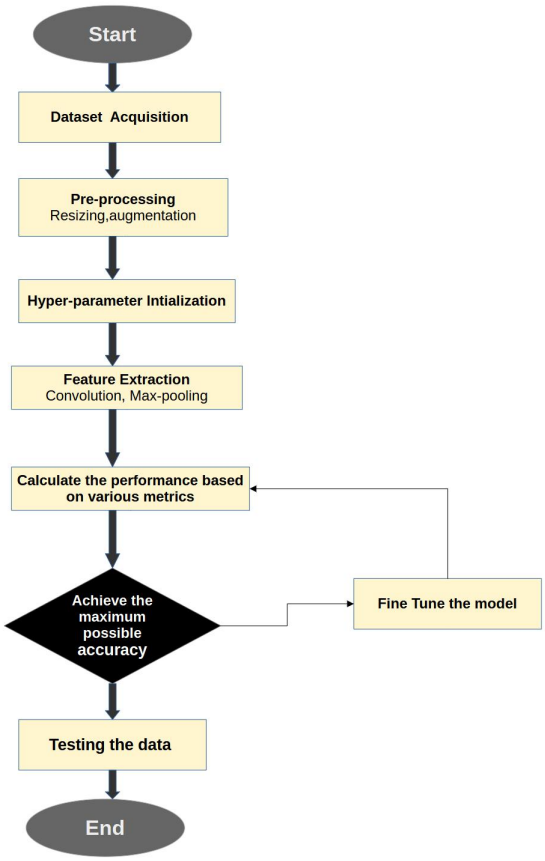


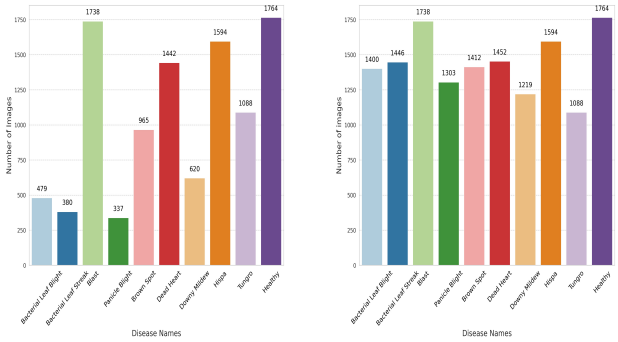
Fig. 1: Workflow of the Proposed Detection Method

function. L1 and L2 regularizations are defined in Equation 1 and 2, respectively. Here $W^{(p)}$ represents the weight matrix of the pth layer in the neural network, $|W|$ is the L1 norm of the weight matrix W, which is the sum of absolute values of all the weights in the matrix and m denotes the total number of weights.

$$\left(W^{(p)}\right)_{l_1} = \|W\| = \sum_{k=1}^{m} |W_k| \tag{1}$$

$$\left(W^{(P)}\right)_{l_2} = \|W\|^2 \tag{2}$$

4) Activation Function: An activation function introduces nonlinearity into a neural network layer. This nonlinearity is essential for the network to learn complex data patterns. At the end of each layer, it determines what information should be passed on to the next neuron. The ReLU activation function is used here with the outputs of convolutional and fully connected layers since it leads to better performance and faster learning. For the output layer, i.e., the last layer of the neural network, the softmax activation function is used to normalize the outputs.



(a) Before Augmentation (b) After Augmentation

Fig. 2: Dataset visulaization

B. Disease Detection by Transfer Learning

In agriculture, collecting a sufficiently labeled dataset is particularly challenging due to factors such as local weather conditions, difficulty distinguishing between plant diseases, and the limited availability of experts. Additionally, overfitting may occur due to the small training dataset, leading to poor classification performance by deep CNN models. One solution is transfer learning, which uses a pre-trained model on a large dataset and applies the acquired knowledge for training on a new dataset. Figure 1 depicts the workflow of the proposed rice leaf disease detection system using transfer learning. The process executes in various steps.

1) Dataset Acquisition: Data collection is the first step in developing our deep learning model. Data are collected from publicly available databases, academic or industry collaborations, or manually acquired data. In this work, we exercise 10407 images from Kaggle. Both healthy and infected leaves are considered.

2) Image Pre-processing: Preprocessing the acquired images enhances the quality and quantity of data. To achieve these objectives, image resizing, normalization, and augmentation techniques are employed. The images are uniformly rescaled to 256 $\times$ 256 pixels to ensure homogeneity in the dataset. Normalization is applied to standardize the pixel values to a range between 0 and 1, which aids in easing the learning process of the model. Furthermore, augmentation techniques, such as rotation, flipping, and shearing, increase the dataset's size and improve the model's generalization capabilities. Figure 2 gives dataset visualization before and after augmentation. The dataset is imbalanced in Figure 2(a), with a significant difference in the number of samples across different classes. After applying augmentation techniques to the dataset, the classes are balanced, as shown in Figure 2(b).

3) Feature Extraction: Feature extraction is performed in the convolutional layers in the proposed model to extract essential features from scaled input images. These features are further processed by a non-linear activation function called rectified linear unit (ReLU) and various pooling operations, reducing the feature maps' dimensionality. Combining convolutional and pooling layers effectively filters the input images to generate informative features in our proposed system.

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4) Classification: The next phase involves classifying the data into multiple classes. Extracted features are utilized in this phase. The classification process is carried out by the fully connected layers of a CNN architecture selected for our model, which maps extracted features to appropriate categories, such as whether the input data is disease-infected. In the last layer of the CNN architecture, the softmax function is applied for classification purposes. In this work, ten classes shown in Figure 3 are adjusted based on the specific requirements of the classification task.

IV. EXPERIMENTAL EVALUATION

This section evaluates the proposed rice leaf disease detection approach and demonstrates its effectiveness. Multiple standard metrics are measured to assess its effectiveness.

A. Dataset Preparation

An open-source database called Kaggle is referred to obtain the dataset. The dataset contains 10407 rice leaf images that are categorized into ten groups. These groups represent various health conditions, including healthy leaves, leaf streak, bacterial blight, blast, tungro, downy mildew, hispa, panicle blight, and brown spot. Each group has a specific number of images. Figure 3 presents their health conditions. To train the selected CNN models, a portion of the dataset is used, while the remaining images are kept aside for testing and validation purposes. We maintain a ratio of 80:20.

B. Performance Metrics

Four performance metrics are used to evaluate our proposed models' effectiveness: accuracy, precision, recall, and f1-score. Accuracy is the primary metric that indicates how accurately the model can detect rice leaf diseases and distinguish between ten classes of leaf images in the dataset. It is calculated using Equation 3, where P represents correctly classified positive samples, N means correctly classified negative samples, F denotes wrongly classified samples as precisely classified ones, and G conveys accurately categorized samples as incorrect ones. By analyzing these metrics, the acceptability of the model can be determined and compared to other models.

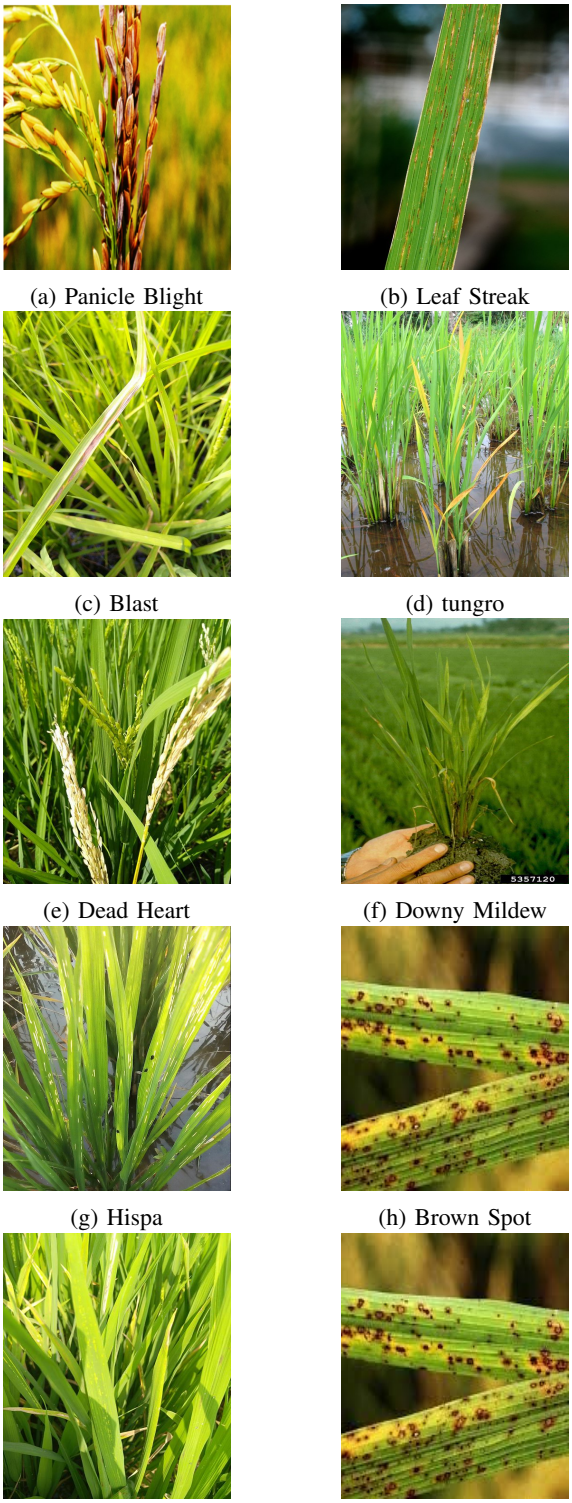
$$accuracy = \frac{P + N}{P + N + F + G} \tag{3}$$

The next performance metric is precision which measures the ability of the model to classify true negatives correctly. It represents the number of positive class predictions that belong to the positive class. The metric is defined in Equation 4.

$$Precision = \frac{P}{P + F} \tag{4}$$

Recall the third performance metric measures the ability of our model to identify infected leaves among all leaves. It is also known as the "true positive rate" of a test. The recall is calculated using Equation 5.

$$Recall = \frac{P}{P + G} \tag{5}$$

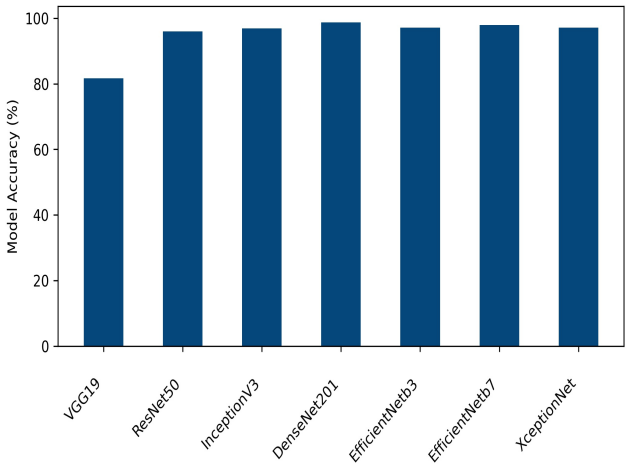


(a) Panicle Blight (b) Leaf Streak (c) Blast (d) tungro (e) Dead Heart (f) Downy Mildew (g) Hispa (h) Brown Spot (i) Leaf Blight (j) Healthy

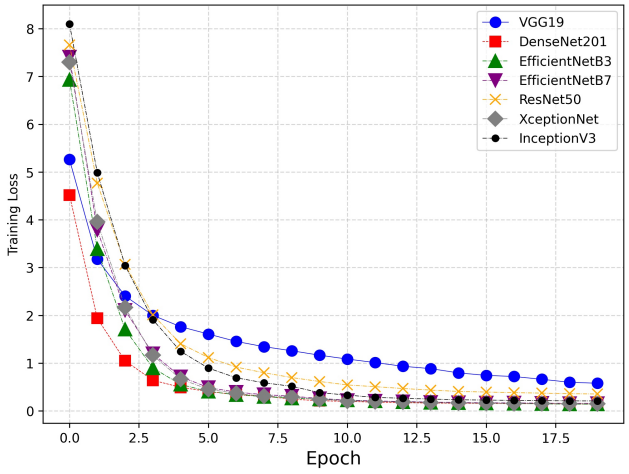
Fig. 3: Health Status of Rice Leaves

F1 score, the final metric, is considered for disease classification. The metric is the harmonic mean of precision and recalls defined in Equation 6. Some experts even prefer it over accuracy, as it provides a more comprehensive evaluation of the model's performance.

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(a) Accuracy



(b) Training Loss

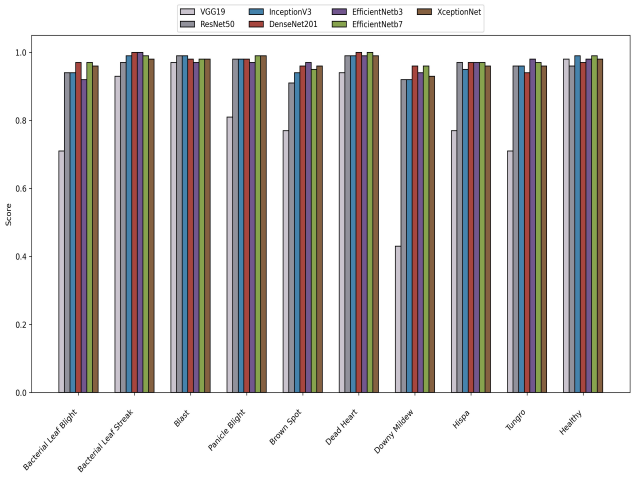
Fig. 4: Accuracy and loss behavior of the pre-trained models

$$F_1 - Score = 2 \times \frac{precision \times recall}{precision + recall} \tag{6}$$

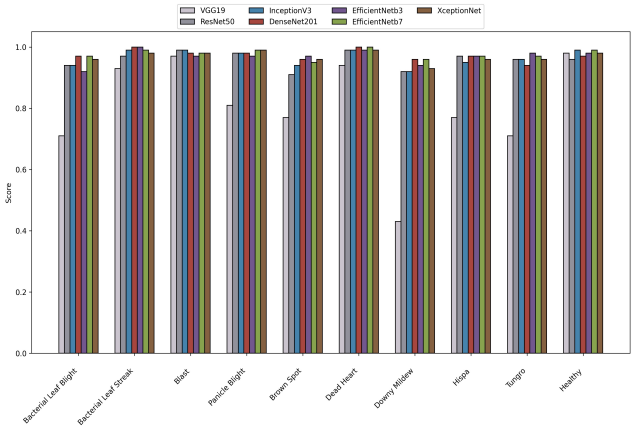
C. Result Analysis

The proposed approach involves evaluating various standard performance metrics using a set of transfer learnings. In particular, seven pre-trained deep learning models are tested and fine-tuned to obtain satisfactory results. These models are VGG19, InceptionV3, EfficientNetB3, EfficientNetB7, XceptionNet, ResNet50, and DenseNet201. The training process conducted on these models utilizes 80% of the available data. The training phase runs for 20 epochs by which the selected models produced promising results. The models are then tested on the remaining 20% of the data.

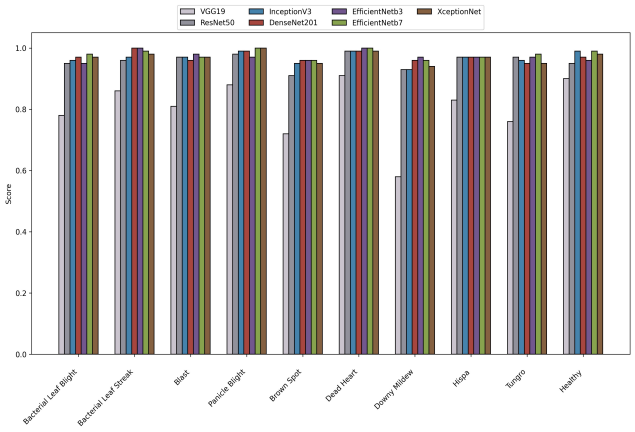
Figure 4 demonstrates the performance of our approach with these models. Accuracy is shown in Figure 4(a). Figure 4(b) shows that the training loss ranges from 0.58 to 0.13 by 20 epochs. The DenseNet201 achieves the lowest training loss, and VGG19 has the highest. Other performance metrics such as precision, recall, and F1-Score are seen in



(a) Precision



(b) Recall



(c) F1 Score

Fig. 5: Performance behavior of the pre-trained models

Figure 5. Regarding precision, DenseNet201, EfficientNetB3, EfficientNetB7, and XceptionNet showed favorable results for all classes, earning a precision score of 100% for certain rice diseases, such as bacterial panicle blight and bacterial leaf streak. VGG19 exhibits the lowest precision score of 0.68 for brown spot disease. Figure 5(a) gives the precision metric our approach achieves. The recall score is seen in Figure 5(b). The DenseNet201 outperforms the other models by achieving

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a 100% recall score for bacterial leaf streak and deadly heart disease and more than 95% for all other disease classes. In Figure 5(c), EfficientNetB3 and EfficientNetB7 achieve a 100% score for bacterial leaf streak, panicle blight, and deadly heart disease, while VGG19 has the lowest score of 0.58 for downy mildew disease.

The training and testing accuracy of the underlined models can be analyzed in Table I. These results demonstrate the effectiveness of the transfer learning approach and the potential of the pre-trained deep learning models for accurate disease detection. DenseNet201 achieves the highest training and testing accuracy of 99.99% and 98.75%, respectively.

TABLE I: Comparison on Model Accuracy

Model Name	**Training Accuracy (%)**	**Testing Accuracy (%)**
VGG19	86.97	81.71
EfficieNetB3	99.98	97.92
EfficieNetB7	99.99	97.63
XceptionNet	98.19	97.16
RseNet50	99.34	95.98
InceptionV3	99.47	96.88
DenseNet201	99.99	98.75

D. Comparision Study

The proposed approach for detecting rice leaf disease shows promising results and outperforms many existing systems. Mahesh *et al.* [12] achieved an accuracy of up to 93.11%, while our model achieves more accuracy by 6.05%. Similarly, Chen *et al.* [2] observed the F1-score for brown spot disease up to 0.91, whereas the metric in our proposed system reaches 0.96, indicating superior performance. Moreover, the method proposed by Yang *et al.* [8] achieved the highest accuracy of 95% with 10-fold cross-validation. It is also more time-consuming due to fewer datasets. In contrast, our proposed system performs better accuracy with less computation on a larger dataset, making it a more efficient and effective method for rice leaf disease detection. Overall, the proposed approach holds significant potential for detecting rice leaf disease accurately and efficiently, outperforming many existing systems.

V. Conclusion

Detecting diseases in rice crops is critical to the agricultural sector, and timely detection can significantly improve crop yields. This paper has developed a novel method for detecting common rice leaf diseases. The technique uses a transfer learning approach based on image analysis. The proposed mechanism has employed seven pre-trained models, namely VGG19, InceptionV3, EfficientNetB3, EfficientNetB7, XceptionNet, ResNet50, and DenseNet201, to analyze and classify images of rice leaves affected by various diseases. The models have been trained on a large dataset, making them robust. The results of the experiments are encouraging. Therefore, the proposed approach demonstrates its ability to classify rice leaf diseases accurately and has the potential to yield better production of rice crops. Future work includes investigating other CNN models for the comprehensive effectiveness of the proposed approach and its extension to cover different parts of a rice plant.

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